

```
#include <iostream>
```

```
#include <vector>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int n, amount;
```

```
    cout << "Enter number of coins: ";
```

```
    cin >> n;
```

```
    vector<int> coins(n);
```

```
    cout << "Enter coins denomination: ";
```

```
    for (int i = 0; i < n; i++)
```

```
        cin >> coins[i];
```

```
    cout << "Enter amount: ";
```

```
    cin >> amount;
```

```
    vector<int> dp(amount + 1, amount + 1);
```

```
    dp[0] = 0;
```

```
    for (int i = 1; i <= amount; i++)
```

```
    {
```

```
        for (int j = 0; j < n; j++)
```

```
        {
```

```
            if (coins[j] <= i)
```

```
    {  
        dp[i] = min(dp[i], dp[i - coins[j]] + 1);  
    }  
}  
}
```

```
if (dp[amount] == amount + 1)  
    cout << "Amount cannot be formed";  
else  
    cout << "Minimum number of coins = " << dp[amount];  
  
return 0;  
}
```